Gaussian Mixture Models

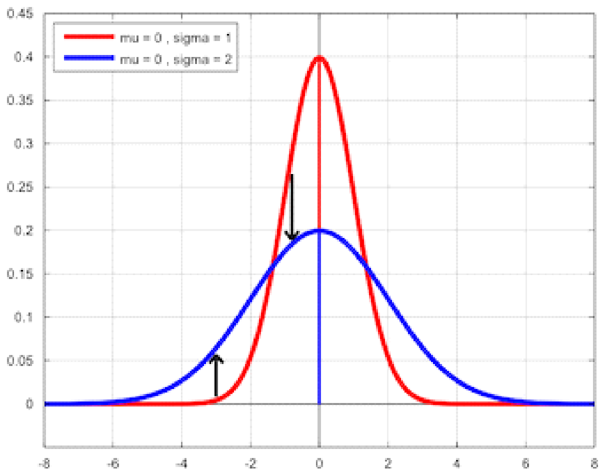
07 October 2025 18:21

- Gaussian Mixture Models (GMMs) are statistical models that represent the data as a mixture of Gaussian (normal) distributions.
- These models can be used to identify groups within the dataset, and to capture the complex, multi-modal structure of data distributions.
- GMMs are used in a variety of machine learning applications, including Clustering, density estimation, and pattern recognition.

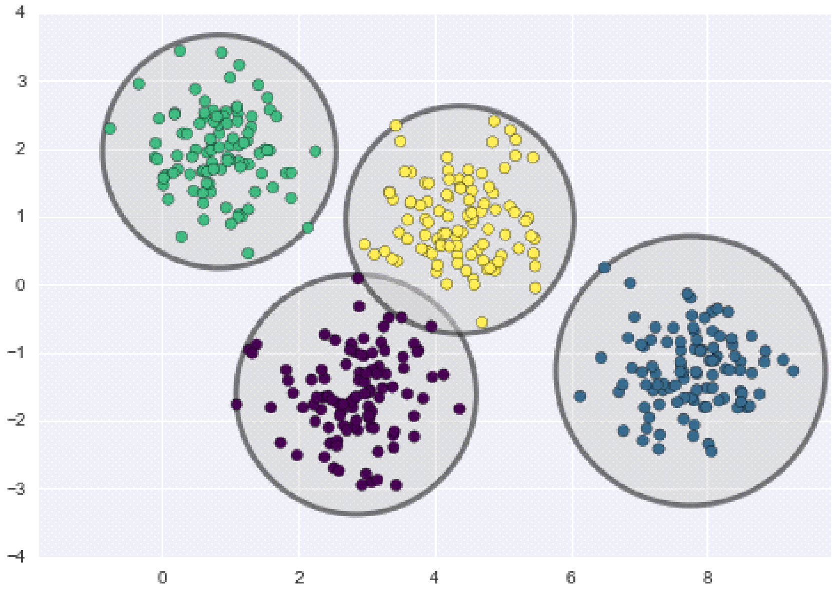
Why use GMM?

Gaussian mixture models can be used to cluster unlabeled data in much the same way as k-means. There are, however, a couple of advantages to using Gaussian mixture models over k-means.

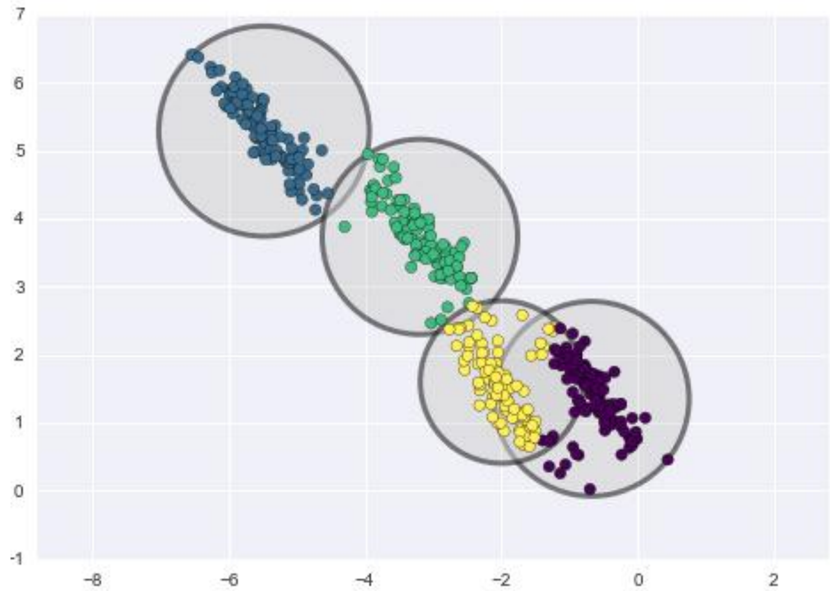
k-means does not account for variance(width of the bell shape curve). In two dimensions, variance/ covariance determines the shape of the distribution.



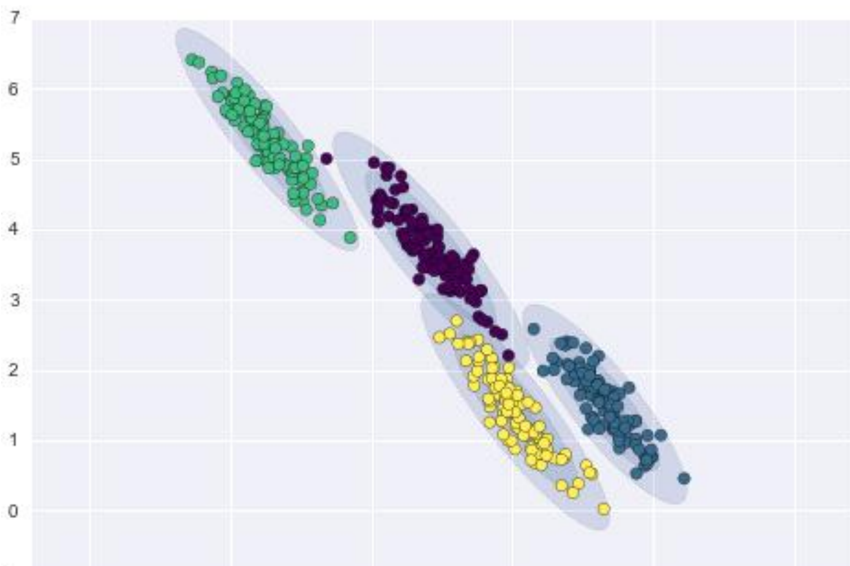
k-means model places a circle (or, in higher dimensions, a hyper-sphere) at the center of each cluster, with a radius defined by the most distant point in the cluster.



- This works fine for when data is circular. However, when data takes on different shape, we end up with something like this.



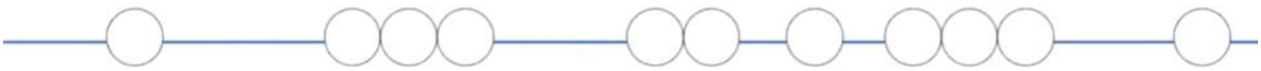
- In contrast, Gaussian mixture models can handle even very oblong clusters.



- K-means performs hard classification whereas GMM performs soft classification, i.e. in k-means, data point is deterministically assigned to one and only one cluster, but in reality there may be overlapping between the cluster GMM provide us the probabilities of the data point belonging to each of the possible clusters.

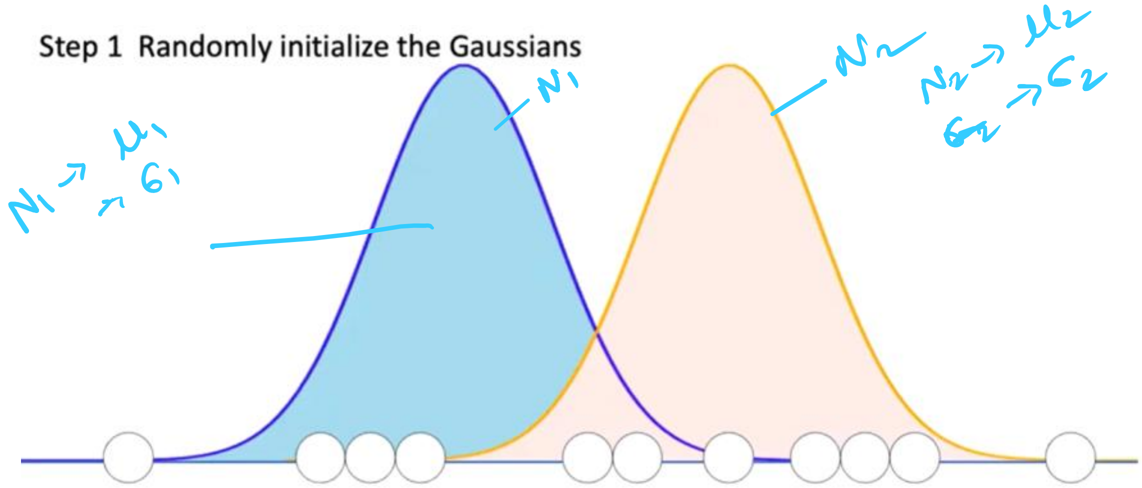
Gaussian Mixture Models :

- **Assumption**: GMMs assume data is generated from a mixture of a fixed number of Gaussian (normal) distributions.

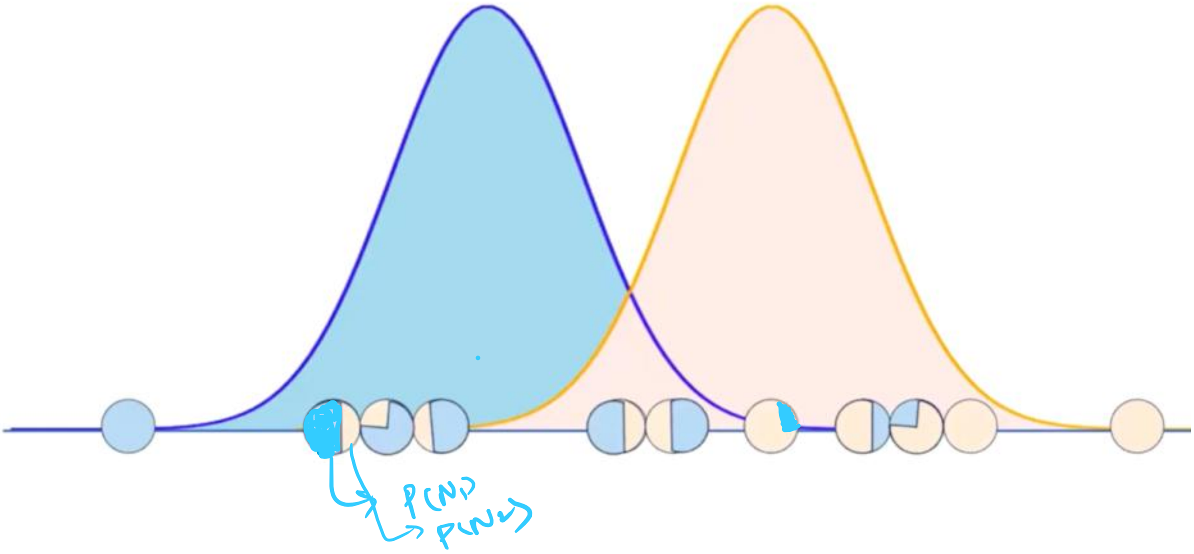


- Needs to decide the number of Gaussian distributions.

Step 1 Randomly initialize the Gaussians

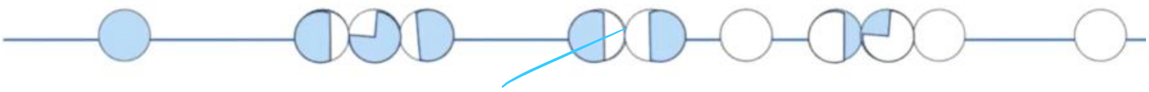


Step 2 Calculate the responsibility(probability of belongingness) for each point

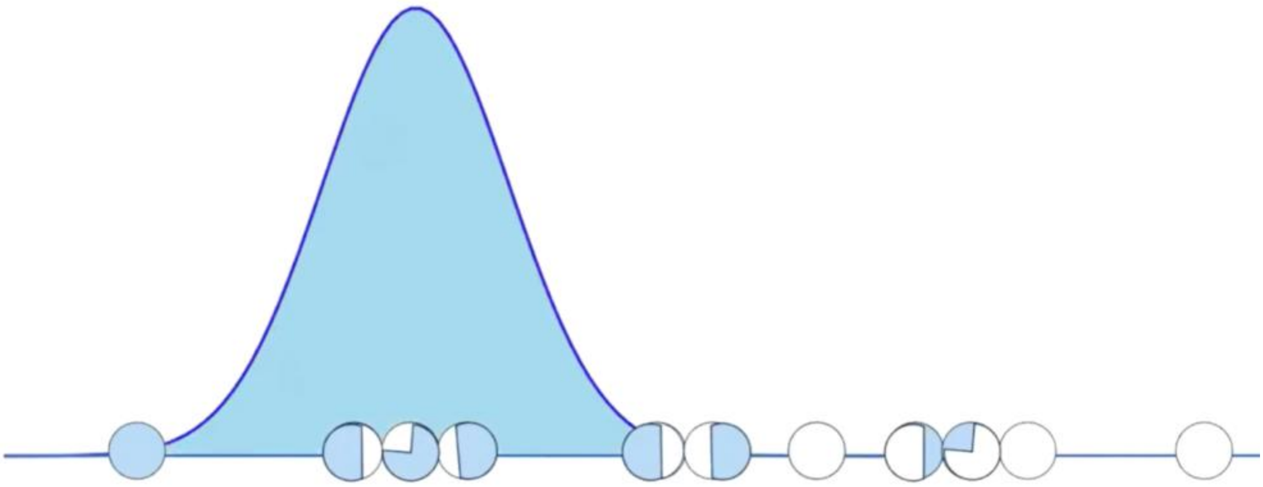


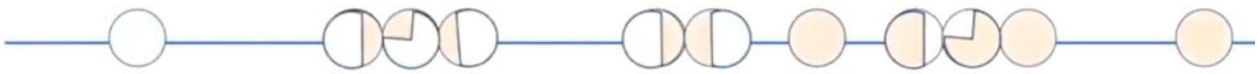
Step 3 Considering the responsibility calculated in the previous step, figure out more appropriate Gaussians

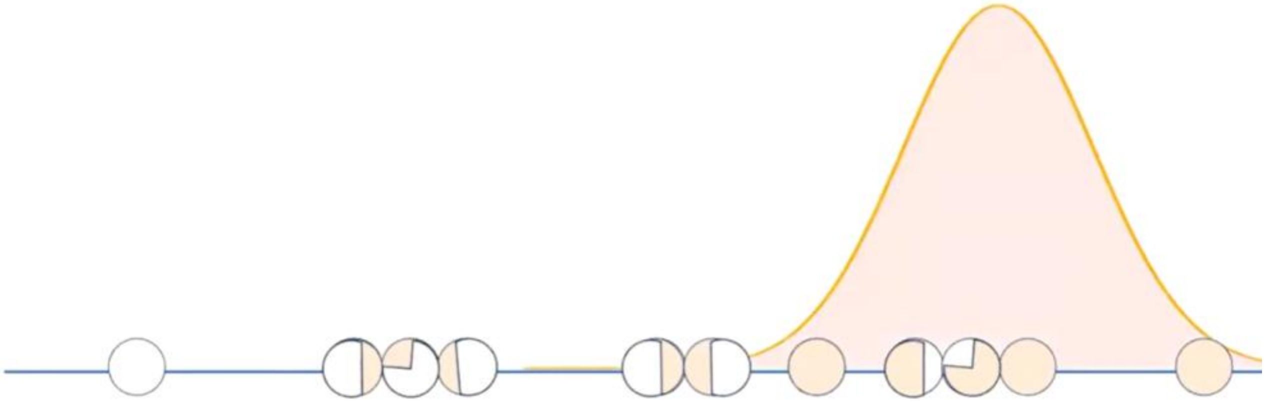
New of μ_1 & σ_1
and μ_2 & σ_2



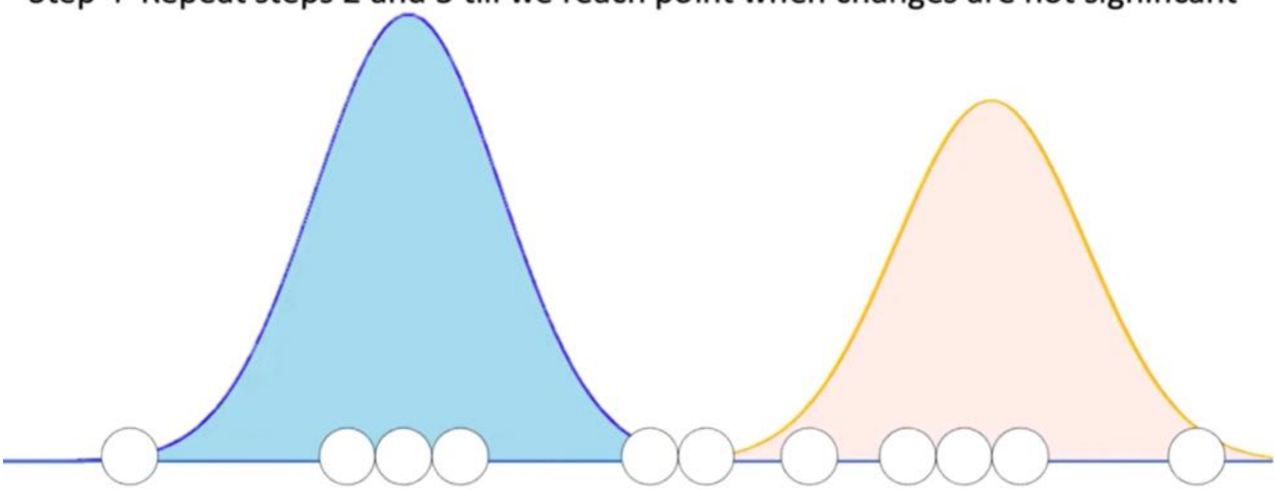
Now, compute the normal distribution again

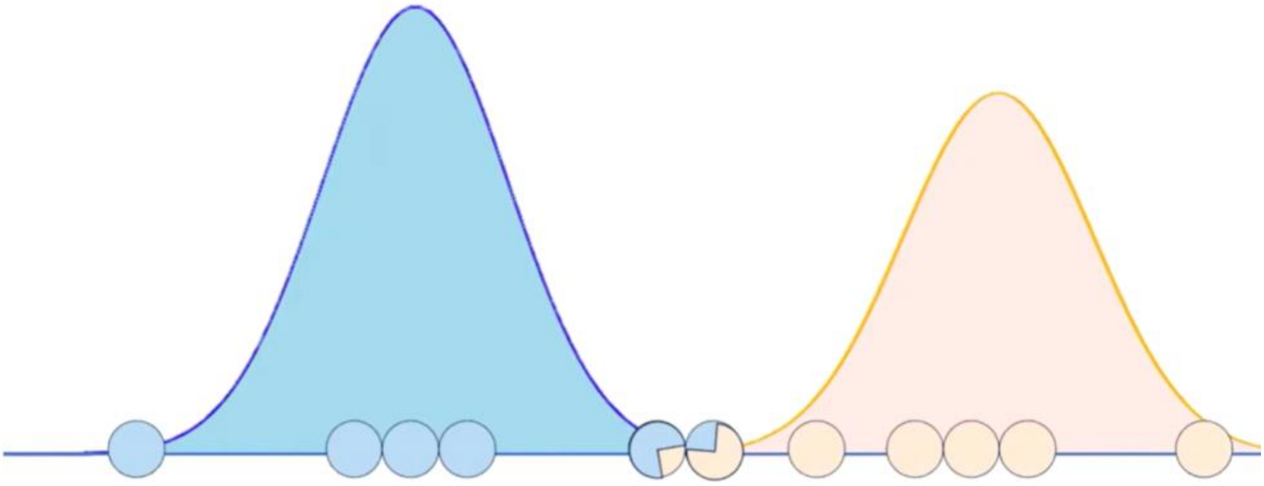




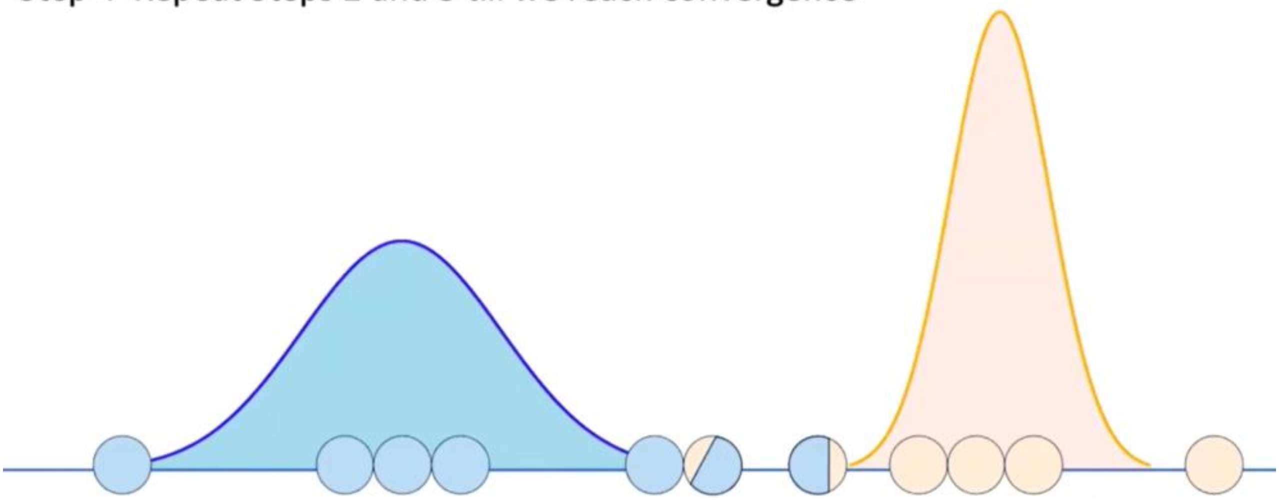


Step 4 Repeat steps 2 and 3 till we reach point when changes are not significant





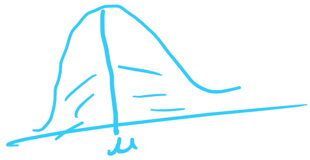
Step 4 Repeat Steps 2 and 3 till we reach convergence

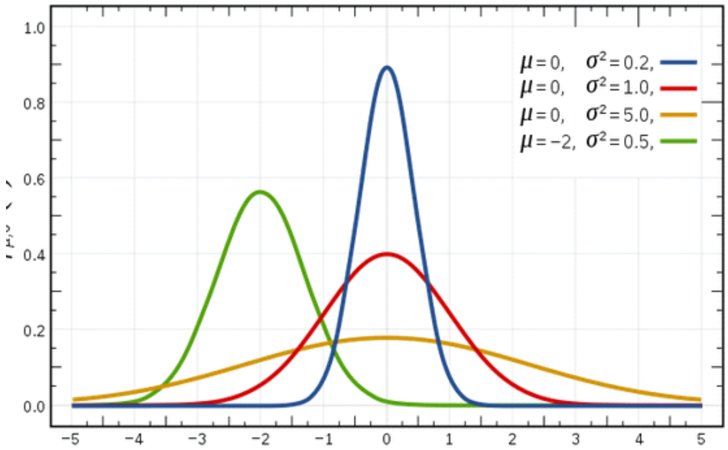


The Gaussian Distribution

02 December 2025 12:50

- It has a bell-shaped curve, with the data points symmetrically distributed around the mean value.
- The image below has a few Gaussian distributions with a difference in mean (μ) and variance (σ 2).
- Remember that the higher the σ value the greater the spread:





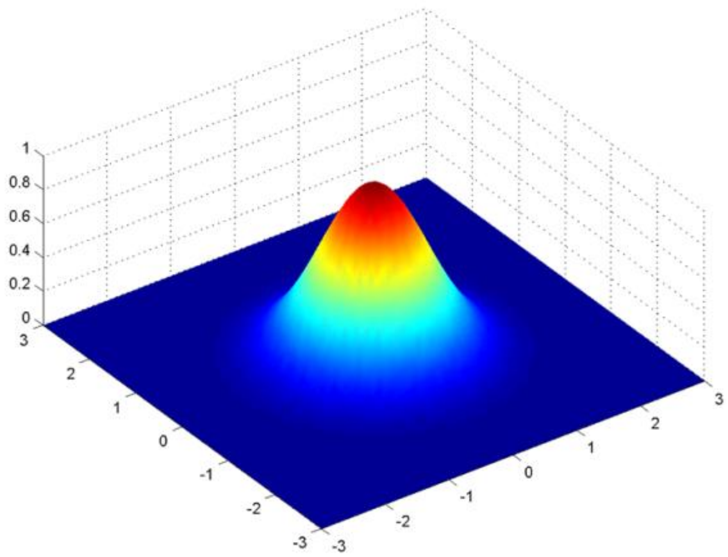
Gaussian / Normal distribution

In a one dimensional space, the probability density function of a Gaussian distribution is given by:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where μ is the mean and σ2 is the variance.

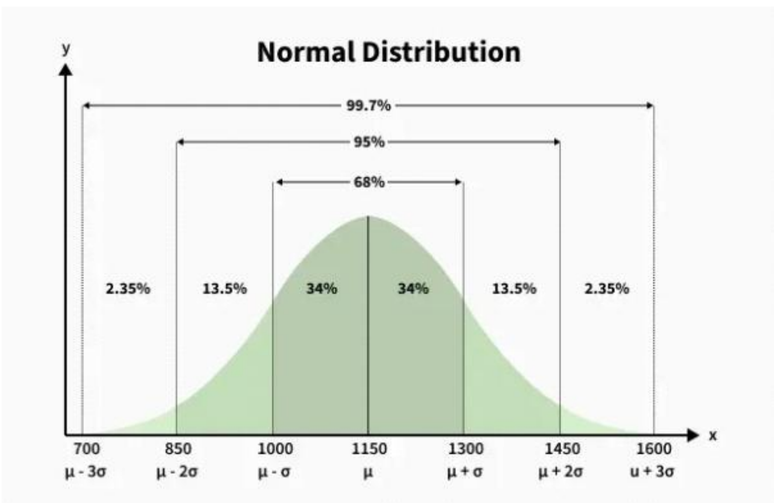
But this would only be true for a single variable. In the case of two variables, instead of a 2D bell-shaped curve, we will have a 3D bell curve as shown below:



The probability density function would be given by:

$$f(x \mid \mu, \Sigma) = \frac{1}{\sqrt{2\pi|\Sigma|}} \exp\left[-\frac{1}{2} (\boldsymbol{x}-\boldsymbol{\mu})^t \Sigma^{-1} (\boldsymbol{x}-\boldsymbol{\mu}) \right]$$

where x is the input vector, μ is the 2D mean vector, and Σ is the 2×2 covariance matrix. The covariance would now define the shape of this curve. We can generalize the same for d-dimensions.



Characteristics of the Normal or Gaussian Distribution

Characteristics of the normal or Gaussian distribution:

- It’s bell-shaped with most values around the average.
- It has only one peak or mode.
- It stretches out forever in both directions.
- It's mean, median, and mode are the same.

Example:

- Its spread is measured by its standard deviation.

x={2,4,5,3,6}

- The total area under its curve equals 1.

We will compute:

1. Mean (μ)
2. Standard deviation (σ)
3. Z-scores
4. Normal PDF value for one number
5. Normal CDF value (using Z-value)

1. Compute the Mean (μ)

$$\mu = \frac{2+4+5+3+6}{5}$$

$$\mu = \frac{20}{5} = 4$$

2. Compute Standard Deviation (σ)

$$\sigma = \sqrt{\frac{1}{5}\sum(x_i - \mu)^2}$$

Compute each:

x_i	$x_i - \mu$	$(x_i - \mu)^2$
2	2 − 4 = −2	4
4	0	0
5	1	1
3	−1	1
6	2	4

Sum:

$$4+0+1+1+4 = 10$$

Then:

$$\sigma = \sqrt{\frac{10}{5}} = \sqrt{2} \approx 1.414$$

3. Z-score for each value

$$z = \frac{x - \mu}{\sigma}$$

Compute:

- For **2**:

$$z = \frac{2 - 4}{1.414} = \frac{-2}{1.414} \approx -1.414$$

- For **4**:

$$z = \frac{4 - 4}{1.414} = 0$$

- For **5**:

$$z = \frac{5 - 4}{1.414} = \frac{1}{1.414} \approx 0.707$$

- For **3**:

$$z = \frac{3 - 4}{1.414} = -0.707$$

- For **6**:

$$z = \frac{6 - 4}{1.414} \approx 1.414$$

✅ 4. Compute the Normal PDF for one value

Let's compute PDF for **x = 5**

Formula:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Plug values:

- μ = 4
- σ = 1.414
- x = 5

Step 1: Compute constant part

$$\sigma\sqrt{2\pi} = 1.414 \times \sqrt{6.283}$$

$$\sigma\sqrt{2\pi} \approx 1.414 \times 2.506 \approx 3.544$$

Thus:

$$\frac{1}{\sigma\sqrt{2\pi}} = \frac{1}{3.544} \approx 0.2823$$

Step 2: Compute exponent

$$(x-\mu)^2 = (5-4)^2 = 1$$

$$2\sigma^2 = 2 \times 2 = 4$$

$$-\frac{1}{4} = -0.25$$

Exponent:

$$\exp(-0.25) \approx 0.7788$$

✅ 5. Normal CDF for x = 5

CDF uses the Z-score:

$$z = 0.707$$

(Using standard normal table or approximation)

$$\Phi(0.707) \approx 0.760$$

Thus:

$$P(X \leq 5) = 0.760$$

Expectation-Maximization

02 December 2025 12:55

What is Expectation-Maximization?

Excellent question!

(μ, σ)

Expectation-Maximization (EM) is a statistical algorithm for finding the right model parameters. We typically use EM when the data has missing values, or in other words, when the data is incomplete.

Broadly, the Expectation-Maximization algorithm has two steps:

- **E-step:** In this step, the available data is used to estimate (guess) the values of the missing variables
- **M-step:** Based on the estimated values generated in the E-step, the complete data is used to update the parameters

Gaussian Mixture Model (GMM) Algo

02 December 2025 13:00

1. What is a Gaussian Mixture Model?

A **Gaussian Mixture Model (GMM)** is a **probabilistic model** that assumes data is generated from a **mixture of several Gaussian distributions** with unknown parameters. GMM can model **elliptical clusters**, overlapping distributions, and soft cluster membership.

2. Mathematical Formulation

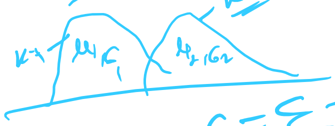
A GMM with *K components* is:

x: 5, μ, σ

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

→ Gaussian

k=2



Where:

- π_k = mixture weights (each ≥ 0, sum to 1)
- μ_k = mean of component k
- Σ_k = covariance matrix of component k

σ = Σ

Each component is a multivariate Gaussian:

$$\mathcal{N}(x|\mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right)$$

GMM is a probability distribution over data points.

3. Soft Clustering

Unlike k-means:

- k-means → **hard assignment** (one cluster)
- GMM → **soft assignment** (probabilities for each cluster)

For a data point x_i, the responsibility of cluster k is:

$$\gamma_{ik} = p(z_k = 1 | x_i)$$

These are posterior probabilities.

4. How GMM is Trained: Expectation–Maximization (EM)

To estimate π_k, μ_k, Σ_k, GMM uses the **EM algorithm**.

Step 1: Initialization

Choose:

- Random means
- Covariances (often identity)
- Equal mixture weights

Step 2: E-Step (Expectation)

Compute cluster probabilities:

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i|\mu_j, \Sigma_j)}$$

Step 3: M-Step (Maximization)

Update parameters.

Effective number of points:

$$N_k = \sum_{i=1}^{N} \gamma_{ik}$$

Update means:

$x_i \in k_1$ and $x_i \in k_2$

$$\mu_k = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} x_i$$

Update covariances:

$$\Sigma_k = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik}(x_i - \mu_k)(x_i - \mu_k)^T$$

Update mixture weights:

$$\pi_k = \frac{N_k}{N}$$

Step 4: Check Convergence

Compare log likelihood:

$$\log p(X) = \sum_{i=1}^{N} \log \left(\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k) \right)$$

Stop when change is small.

5. Choosing Number of Components (K)

Common methods:

- **Bayesian Information Criterion (BIC)**

$$\mathrm{BIC} = -2 \log L + p \log N$$

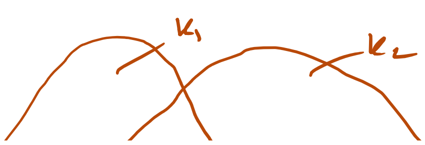
- **Akaike Information Criterion (AIC)****

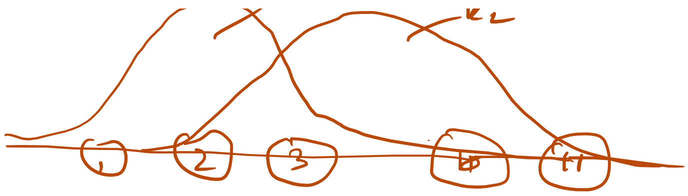
$$\mathrm{AIC} = -2 \log L + 2p$$

Lower values = better.

Example $x = \{1, 2, 3, 10, 11\}$

cluster 1 → small value (1, 2, 3)
cluster 2 → large value (10, 11)





① Random Initialization of model parameter

for k_1: $\mu_1 = 2$, $\sigma_1 = 1$ | for k_2: $\mu_2 = 10.5$, $\sigma_2 = 1$

$\pi_1 = 0.5$ $\quad\quad$ $\pi_2 = 0.5$

2. E-step (compute responsibilities)

k_1: $$\gamma_{i1} = \frac{\pi_1 \mathcal{N}(x_i \mid \mu_1, \sigma_1)}{\pi_1 \mathcal{N}(x_i \mid \mu_1, \sigma_1) + \pi_2 \mathcal{N}(x_i \mid \mu_2, \sigma_2)}$$

k_2: $$\gamma_{i2} = 1 - \gamma_{i1}$$

$x = 1$

$$\gamma_{11} = \frac{0.5 \times \left[\underbrace{\frac{1}{\sqrt{2\pi} \times 1} \times e^{-\left(\frac{(1-2)^2}{2\times 1}\right)}}_{0.242}\right]}{0.5 \times (\quad) + 0.5 \times \underbrace{\left[\frac{1}{\sqrt{2\pi} \times 1} \times e^{-\left(\frac{(1-10.5)^2}{2\times 1}\right)}\right]}_{0}}$$

$$\gamma_{11} = \frac{0.5 \times 0.242}{0.5 \times 0.242 + 0}$$

$$\gamma_{11} = 1$$

$$\gamma_{12} = 1 - 1 = 0$$

x_i $\quad\quad$ γ_{i1} $\quad\quad$ γ_{i2}

1	1	0
2	1	0
3	1	0
10	0	1
11	0	1

④ M- stem (updates the parameters)

$k_1 : \mu_1, \sigma_1, \pi_1$

$$\mu_1 = \frac{1 \times r_{11} + 2 r_{12} + 3 \times r_{13}}{3}$$

$$\mu_1 = \frac{1 \times 1 + 2 \times 1 + 3 \times 1}{3}$$

$$= \frac{6}{3} = 2$$

$$\sigma^2 = \sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}$$

$$\sigma^2 = \frac{(1-2)^2 + (2-2)^2 + (3-2)^2}{3}$$

$$= 2/3$$

$k_2 : \mu_2, \sigma_2, \pi_2$

$$\mu_2 = \frac{10 \times 1 + 11 \times 1}{2}$$

$$\mu_2 = \frac{21}{2} = 10.5$$

$$\sigma^2 = \frac{(10-10.5)^2 + (11-10.5)^2}{2}$$

$$\sigma^2 = \frac{5}{2}$$

$$\pi_1 = \frac{1}{N} \sum_{i=1}^{N} r_{ik}$$

$$\pi_1 = \frac{1}{5}(1 + 1 + 1 + 0 + 0)$$

$$\pi_2 = \frac{1}{5}(0 + 0 + 0 + 1 + 1)$$

$$\pi_2 = \frac{2}{5}$$

$$= 0.4$$

$\pi_1 = \frac{1}{5}($

$= \underline{\underline{0.4}}$

$\pi_1 = \frac{3}{5} = 0.6$

1, 2, 3 → Prob to belong k_1 is 60 %.

1, 2, 3 → Prob to belong k_2 is $\underline{\underline{40}}$ %.

10, 11 → Prob to belong k_2 → 60

10, 11 → — " ——— " — k_1 → 40

covariance matrix

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A **covariance matrix** shows how multiple variables vary **together**.

- The diagonal elements are the **variances** of each variable.
- The off-diagonal elements are the **covariances** between pairs of variables.

If covariance > 0 → variables increase together

If covariance < 0 → one increases while the other decreases

If covariance = 0 → no linear relationship

Numerical Example

Suppose we have two variables:

X = [2, 4, 6]

Y = [1, 3, 5]

Step 1: Compute means

$$\bar{X} = \frac{2+4+6}{3} = 4$$

$$\bar{Y} = \frac{1+3+5}{3} = 3$$

Step 2: Compute covariance

$$\text{Cov}(X,Y) = \frac{1}{n-1}\sum(X_i - \bar{X})(Y_i - \bar{Y})$$

$$(2-4)(1-3) = (-2)(-2) = 4$$
$$(4-4)(3-3) = 0$$
$$(6-4)(5-3) = (2)(2) = 4$$

$$\text{Cov}(X,Y) = \frac{4+0+4}{2} = 4$$

Step 3: Compute variances

$$\mathrm{Var}(X) = \frac{(-2)^2 + 0^2 + 2^2}{2} = \frac{8}{2} = 4$$

$$\mathrm{Var}(Y) = \frac{(-2)^2 + 0^2 + 2^2}{2} = \frac{8}{2} = 4$$

Covariance Matrix

$$\Sigma = \begin{bmatrix} \mathrm{Var}(X) & \mathrm{Cov}(X, Y) \\ \mathrm{Cov}(X, Y) & \mathrm{Var}(Y) \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

Interpretation

- Both variables have variance = 4
- Covariance = 4 (strong positive relation → as X increases, Y increases)